

Component library — the canonical reusable component catalog

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Master technical specification — Volume 10 (Design System), nano-detail deep-dive. This document **owns** HelixVPN's complete reusable **component library** — every component the `helix_design` submodule ships, used across the three apps (Client, Console, Connector) and eight platforms. For **every** component it specifies: purpose; anatomy (parts); the full state set; variants and sizes; the exact **component tokens** it consumes (cited against the `comp.<component>.<part>.<state>` naming owned by [`design-tokens.md`]); accessibility (roles, focus order, contrast floor, ≥ 44 px hit target, screen-reader labels); **light + dark** rendering; the **no-overlap / no-label-overlay** guarantee (§11.4.162); and an ASCII or Mermaid sketch.

SPEC-ONLY. It describes *what each component is and how it behaves* — not the shipping `helix_design/helix_ui` build. Concrete token **values** live in [`color-system.md`] (hex + contrast) and [`design-tokens.md`] (scales, schema); this doc describes which **slots** each component has and which tokens fill them.

Position. `helix_design` is a fully decoupled, reusable submodule (`vasic-digital/helix_design`, snake_case flat per §11.4.28/.29/.74) that configures **OpenDesign** (the mandatory token/theme engine, §11.4.162). Each component here is **OpenDesign-token-driven, decoupled, reusable**, ships a **light AND dark** variant, and is covered by visual-regression tests ([`visual-regression-and-qa.md`], §11.4.162/.168). No component hard-codes a colour, length, or duration — every value is a token reference.

Boundary with sibling docs. Owns: the component catalog (anatomy, states, slots, allly, sketches). Consumes: the 3-tier token model + comp.* naming [design-tokens.md §1, §2, §7]; the colour roles + connection-state palette

- contrast proofs [color-system.md §2, §3, §4]; the **7-variant ffi::TunnelStatus** the ConnectButton/StatusChip render [v04-client/ffi-surface.md §3.2]; the Shields/ExitOption DTOs [FFI §2]; the client UI overview [04_CLIENT §7]. Layout/screen composition is [screens-client.md]/[screens-console.md]; per-platform adaptation is [platform-adaptation.md]; type/icon scales are [typography-iconography-motion.md].

Evidence base. [DT §N] = final/v10-design/design-tokens.md; [COLOR §N] = final/v10-design/color-system.md; [FFI §N] = final/v04-client/ffi-surface.md; [04_CLIENT §N] = final/03-client-core-and-ui.md; [SPINE §N] = final/SPECIFICATION.md. The component catalog itself is **original HelixVPN design work** (modeled on the well-established headless-component / WAI-ARIA-Authoring-Practices pattern, cited in Sources). Claims not grounded in the evidence base or in this document's own design choices are tagged UNVERIFIED per §11.4.6 — never fabricated.

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0. Component inventory

The canonical catalog. **Component** = the reusable widget; **Used-by** = which of the three apps render it (C = Client, K = Console, N = Connector); **Key states** = the load-bearing states (full sets in each section); **Tokens** = the root component-token namespace it owns (comp.<name>.*, [DT §2, §7]).

Component	Used-by	Key states	Token root
ConnectButton	C	Disconnected · Connecting · Handshaking · Connected{direct relay} · Reconnecting · Down · Danger (× hover/pressed/ focus/disabled)	comp.connectButton.*
StatusChip	C, N	transport-kind × path(direct relay) × rtt-band; idle / connecting / down / danger	comp.statusChip.*

Component	Used-by	Key states	Token root
ServerListItem	C, K	default · hover · pressed · selected · focused · disabled · favourite · loading-rtt	comp.serverListItem.*
ServerList	C, K	populated · filtered · empty · loading(skeleton) · error	comp.serverList.*
ExitPicker	C	closed · open · searching · multihop-chain · selected	comp.exitPicker.*
ShieldToggle	C, N	off · on · pending · disabled · locked-by-policy	comp.shieldToggle.*
MultiHopChainEditor	C, K	1-hop · 2-hop · add-hop · reorder · invalid-chain	comp.hopChain.*
PeerCard / DeviceListItem	K, N	online · idle · offline · expiring-key · revoked · pending-approval	comp.peerCard.*
PolicyEditor primitives (RuleRow, AllowDenyToggle, TargetPicker)	K	allow · deny · draft · conflict · disabled	comp.policyRule.*
TopologyGraph (node, edge)	K	node{healthy degraded down selected} · edge{direct relay blocked}	comp.topology.*
AuditLogRow	K	info · success · warn · error · security; expanded · selected	comp.auditRow.*
Button (primary/secondary/text/danger)	C, K, N	enabled · hover · pressed · focus · disabled · loading	comp.button.*

Component	Used-by	Key states	Token root
TextField / FormField	C, K, N	empty · focused · filled · error · disabled · readonly	comp.field.*
Toggle/Switch	C, K, N	off · on · focus · disabled	comp.switch.*
Checkbox / Radio	C, K, N	unchecked · checked · indeterminate(cb) · focus · disabled	comp.checkbox.* / comp.radio.*
Dropdown / Select	C, K, N	closed · open · option{hover selected disabled} · focus	comp.select.*
Dialog / Sheet / Modal	C, K, N	enter · shown · exit · scrim	comp.dialog.*
Snackbar / Toast	C, K, N	info · success · warn · error; entering · shown · leaving	comp.toast.*
Banner / Alert (incl. Danger banner)	C, K, N	info · success · warning · error · danger	comp.banner.*
Tabs	C, K, N	tab{default selected hover focus disabled}	comp.tabs.*
NavRail / BottomNav / Drawer	C, K, N	item{default selected hover focus} · collapsed/expanded	comp.nav.*
Card	C, K, N	default · interactive(hover/pressed) · selected · disabled	comp.card.*
ListTile	C, K, N	default · hover · pressed · selected · disabled	comp.listTile.*
Badge	C, K, N	neutral · info · success · warning · error · count	comp.badge.*

Component	Used-by	Key states	Token root
Tooltip	C, K, N	hidden · shown	comp.tooltip.*
ProgressIndicator	C, K, N	loading (indeterminate spinner) · connecting (state pulse) · determinate(bar)	comp.progress.*
EmptyState	C, K, N	empty · empty-filtered · error	comp.emptyState.*
Skeleton	C, K, N	shimmer · static (reduce-motion)	comp.skeleton.*

Every row ships **light** + **dark** variants and is covered by a per-theme golden-screenshot in the visual-regression suite ([visual-regression-and-qa.md], §11.4.162). A new component **MUST** appear in this table before it ships (the table is the catalog source-of-truth; the gate CM-component-inventory in [visual-regression-and-qa.md] asserts every comp.* root has a row).

1. Cross-cutting conventions

These conventions apply to **every** component and are stated once here so each component section can reference them instead of repeating.

1.1 Anatomy vocabulary

Each component is described by its **parts** (the named, addressable sub-slots a component token can target). The shared part vocabulary:

Part	Meaning	Typical token leaf
container / bg	the component's background fill	comp.<c>.bg[.<state>]
surface	a raised inner surface (card body)	comp.<c>.surface
border	the outline / hairline	comp.<c>.border[.<state>]
label	the primary text	comp.<c>.label[.<state>]

Part	Meaning	Typical token leaf
supporting	secondary/ caption text	comp.<c>.supporting
icon / leading / trailing	glyph slots	comp.<c>.icon[.<state>]
focusRing	the keyboard- focus indicator	comp.<c>.focusRing (= color.semantic.border.focus)
ripple / overlay	the press/hover state overlay	comp.<c>.overlay.<hover pressed>
radius / padX / padY / gap	shape & spacing	comp.<c>.radius, ...padX, ...gap

1.2 The universal interactive-state model

Every interactive component resolves a **base** appearance modulated by an **interaction layer** (a translucent overlay token), so states compose without re-declaring colours:

```
final = base(component, semantic-state) ⊗ interaction-  
overlay(hover|pressed|focus|disabled)
```

Interaction	Overlay token	Effect
default	none	base
hover	comp.<c>.overlay.hover ($\approx$ on-surface @ 8 % α)	subtle lighten/ darken
pressed	comp.<c>.overlay.pressed ($\approx$ on-surface @ 12 % α) + motion.semantic.press (100 ms)	press feedback
focus (keyboard)	comp.<c>.focusRing = border.focus (2 px, ≥ 3.0 contrast, [COLOR §4.5])	always- visible ring
disabled	base at reduced opacity (opacity.semantic.disabled ≈ 0.38) + no overlay, pointer-events off	inert
loading	base + comp.progress.spinner replacing the label/icon, aria-busy=true	busy

Focus is **never** conveyed by colour alone — the ring is a 2 px outline offset from the container so it is visible against any state fill (focusRing clears the 3.0 non-text floor in both themes, [COLOR §4.5]).

1.3 Accessibility baseline (every component)

- **Role** — each component declares an explicit ARIA/semantics role (stated per section). Flutter Semantics widgets carry the equivalent on iOS/Android/ desktop; web emits the ARIA role; HarmonyOS/Aurora map to their ally trees.
- **Name** — every interactive element has a non-empty accessible name (from visible label, or `semanticLabel` when icon-only). Icon-only controls **MUST** supply an explicit label (e.g. `ConnectButton` announces "Connect" / "Connected, direct path, 28 ms").
- **Focus order** — logical, follows reading order (top→bottom, leading→trailing); modals trap focus; Esc closes dismissibles; tab order never jumps.
- **Hit target** — every interactive target is **≥44 × 44 px** (the touch floor, WAI/Apple HIG/Material). Visually-smaller controls (a 20 px checkbox) keep a 44 px **transparent** tap area centred on the glyph. This is a hard gate (CM-hit-target-44 in [visual-regression-and-qa.md]).
- **Contrast** — text/icon vs its background meets AA-normal 4.5 (or AA-large 3.0 for ≥24 px); non-text UI parts meet 3.0; all proven in [COLOR §4].
- **Reduce-motion** — any looping/transition animation degrades to its static end-state via `motion.semantic.reducedMotion` ([DT §6.4]); nothing strobes (§11.4.107 no-flash).
- **Colour-independence** — state is never conveyed by colour alone; every coloured state also carries an icon and/or text ([COLOR §0]).

1.4 Light + dark by construction

No component declares a theme fork. Every `comp.*` token references a **semantic** token, and the semantic tier owns the {light, dark} fork ([DT §1, §5.1]). Therefore **every component is dual-theme automatically** — swapping `MaterialApp(theme:, darkTheme:)` re-resolves every slot. Each section below notes any **theme-specific behaviour** beyond the automatic colour swap (e.g. elevation shadows darken on dark surfaces, [DT §6.3]; the `ConnectButton` glow is brighter on dark).

1.5 The no-overlap / no-label-overlay guarantee

§11.4.162: "elements **MUST NOT** overlap or overlay labels." Each component guarantees this structurally (layout half; the colour half is [COLOR §5]):

1. **Labels live in their own box.** Every text slot is laid out in a flex/grid cell with intrinsic min-size; it is never absolutely positioned over a sibling. Truncation is *ellipsis* + a tooltip (§11.14), never overflow that collides with a neighbour.

2. **Minimum gaps.** Adjacent interactive/coloured elements keep $\geq \text{space.scale.2}$ (8 px) between them; leading/trailing icons keep $\geq \text{space.scale.1}$ (4 px) from the label.
3. **Overlays use a scrim + their own surface.** Floating parts (tooltip, dropdown menu, dialog) render on an **opaque** surface above an `overlay.scrim`, never as a translucent wash over content ([COLOR §5.2]).
4. **Badges/indicators clear the content.** A count badge or status dot anchored to a corner reserves its own padding box; it never paints over glyph or text.
5. **Visual-regression proof.** Each component's golden screenshot (per theme, per size) is OCR-/overlap-checked: any text bounding-box intersection or sub-3.0 label contrast fails the build (§11.4.162/.168, [visual-regression-and-qa.md]).

1.6 Size & density

Components offer a small closed size set keyed to the spacing scale ([DT §6.1]):

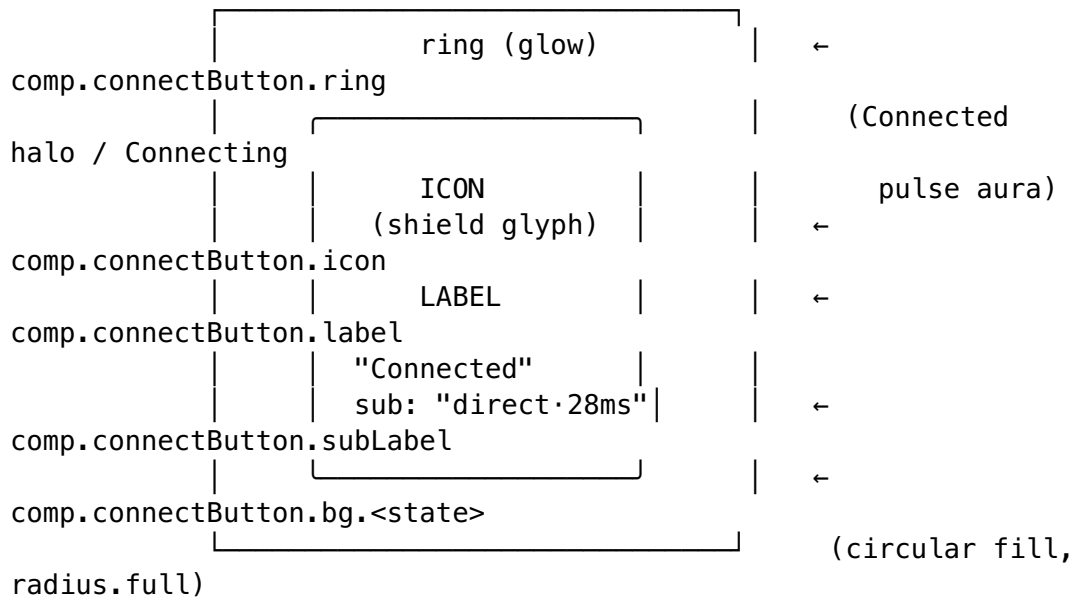
Size	Min height	Pad-X	Use
sm	32 px	<code>space.scale.2</code> (8)	dense tables, Console rows
md (default)	40 px	<code>space.scale.3</code> (12)	most controls
lg	48 px	<code>space.scale.4</code> (16)	primary actions, touch-first Client
xl	56 px	<code>space.scale.5</code> (24)	TV-leanback focus targets

The **tap target** is always ≥ 44 px even when the visual `sm` height is 32 px (transparent padding, §1.3). TV-leanback ([SPINE §3]) forces `xl` and a larger focus ring.

2. ConnectButton — the hero

Purpose. The single most important control in the product: the large circular Connect/Disconnect button that **renders all seven `ffi::TunnelStatus` variants** [FFI §3.2] with the connection-state palette [COLOR §3] and the state motion [DT §6.4]. It is the primary safety signal — the user's "am I protected?" answer at a glance.

Anatomy.



Parts: ring (animated halo), bg (circular fill, the state colour), icon (state glyph), label (state verb), subLabel (path · rtt, only when Connected), focusRing, overlay.<hover|pressed>.

The full state set (all 7 FFI variants × interaction). The bg, icon, label, ring, and motion are all driven by the active `ffi::TunnelStatus`:

<code>ffi::TunnelStatus</code>	<code>comp.connectButton.bg</code>	icon	label	subLabel
Disconnected	<code>...bg.disconnected = action.primary (brand, idle)</code>	shield-off	"Disconnected" / "Tap to connect"	—
Connecting	<code>...bg.connecting = state.connecting.fill (amber)</code>	spinner (state-pulse, §11.15)	"Connecting..."	—
Handshaking	<code>...bg.handshaking (amber, = Connecting)</code>	key-exchange	"Securing..."	—
Connected{direct}	<code>...bg.connected = state.connected.fill (green)</code>	shield-check	"Connected"	"direct · {rtt} ms"
Connected{relay}	<code>...bg.connectedRelay (teal-green)</code>	shield-check-relay	"Connected"	"relay · {rtt} ms"
Reconnecting	<code>...bg.reconnecting (amber, pulsing)</code>	reconnect-arrows	"Reconnecting..."	—
Down{reason}	<code>...bg.down = state.down.fill (orange)</code>	alert-triangle	"Connection lost"	reason (truncated)

<code>ffi::TunnelStatus</code>	<code>comp.connectButton.bg</code>	<code>icon</code>	<code>label</code>	<code>subLabel</code>
<code>Danger{kind}</code>	<code>...bg.danger = state.danger.fill (red)</code>	<code>shield-alert</code>	<code>"Not protected"</code>	<code>kind ("leak" / "kill-switch")</code>

Interaction overlays (§1.2) compose on top: hover lightens, pressed adds the 12 % overlay + 100 ms press scale, disabled (e.g. mid-Handshaking when a tap would be a no-op) drops to 0.38 opacity and announces aria-disabled.

Tokens consumed (all reference semantics, [DT §7, §8]):

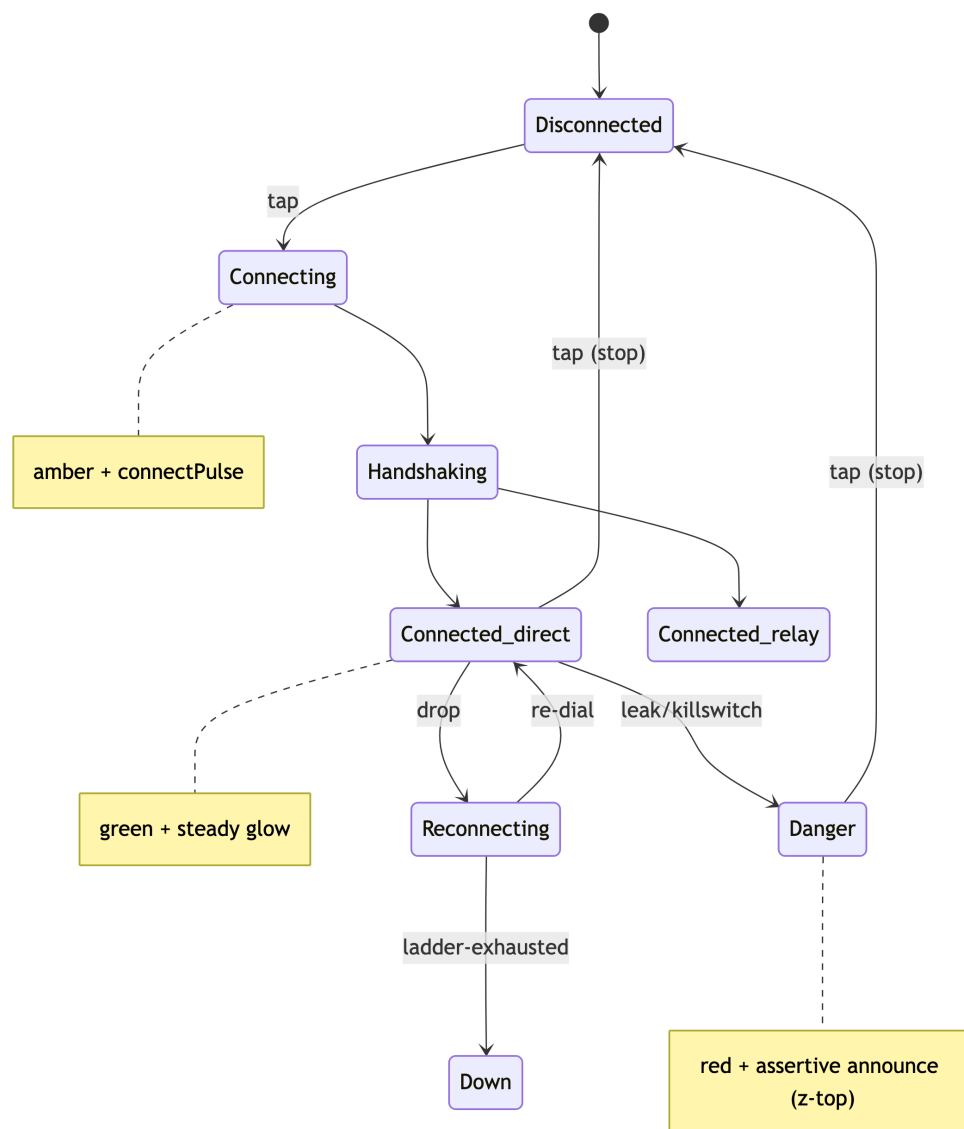
`comp.connectButton.bg`.
`{disconnected, connecting, handshaking, connected, connectedRelay, reconnecting, down, danger}`, `comp.connectButton.label` (= `text.onState`, white on every fill), `comp.connectButton.subLabel` (= `text.onState @ 0.85 α`), `comp.connectButton.icon` (= `text.onState`), `comp.connectButton.ring` (= `state.connected.glow / state.connecting.glow`), `comp.connectButton.radius` (= `radius.scale.full`), `comp.connectButton.size` (= `size.connectButton.diameter`, lg≈180 px / xl≈220 px TV), `comp.connectButton.focusRing` (= `border.focus`), `motion: motion.semantic.connectPulse`, `motion.semantic.stateXfade` (180 ms cross-fade between any two state fills, [DT §6.4]).

Accessibility.

- Role: button. Accessible name is **stateful** and read aloud on change via a polite live-region: "Connecting", "Connected, direct path, 28 milliseconds", "Not protected, leak detected" (the label + subLabel composed). The icon is decorative (aria-hidden) because the name carries the meaning.
- Connecting/Handshaking/Reconnecting set `aria-busy=true`.
- Danger additionally fires an **assertive** announcement (interrupts) because the user must know immediately they are exposed.
- Hit target: the circular target is ≥180 px (far above 44). Keyboard: Space/Enter toggles; focus ring is a 4 px offset ring (`border.focus`).
- Colour-independence: every state has a distinct **glyph** + **label**, so a colour-blind user reads "Connected" vs "Not protected" without the green/red.
- Reduce-motion: the Connecting/Reconnecting pulse becomes a **static** amber ring; the spinner becomes a static arc with an animated `aria-busy` only — it never strobes (§11.4.107).

Light + dark. bg/label/icon swap automatically via the semantic fork ([COLOR §3.1]); on **dark** the ring glow is rendered brighter (the .500 state stops) and the connected glow has a higher alpha so the halo reads on #151920. The white label is white in both themes (it sits on a coloured fill, proven AA in both, [COLOR §4.3]).

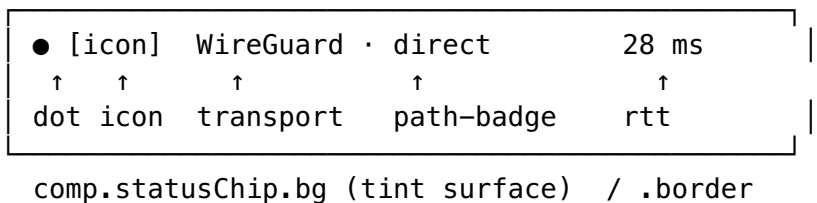
No-overlap guarantee. Label + subLabel are stacked in a centred column with fixed line-boxes; the icon sits above them with a space.scale.2 gap — the glyph never overlays the text. The ring is drawn **outside** the fill radius (a separate larger circle) so it never occludes the label. subLabel truncates with ellipsis + the full value in the accessible name; it never wraps onto the ring.



3. StatusChip

Purpose. A compact, always-visible chip that surfaces the **transport kind**, the **path** (direct / relay), and the **rtt** of the active connection — the "how am I connected" detail that complements the ConnectButton's "am I connected". Rendered on the Client home and on the Connector status surface. Reads from the same `ffi::TunnelStatus::Connected { transport, path, rtt_ms }` [FFI §3.2].

Anatomy.



Parts: dot (state colour dot), icon (transport glyph), label (transport kind from `Transport::kind()`), pathBadge (a sub-pill: "direct"/"relay"), rtt (number + "ms", colour-banded), bg (low-alpha tint of the state colour), border.

States.

State	Source	dot / tint	path-badge	rtt band
connected · direct	Connected{path:"direct"}	green tint, green dot	"direct" (green)	rtt colour band
connected · relay	Connected{path:"relay"}	teal tint, teal dot	"relay" (teal)	rtt colour band
connecting	Connecting/Handshaking	amber tint	hidden	"—"
reconnecting	Reconnecting	amber tint (pulse dot)	hidden	"—"
down	Down	orange tint	hidden	hidden
danger	Danger	red tint	hidden	hidden
idle	Disconnected	grey tint	hidden	hidden

rtt colour band (`comp.statusChip.rtt.<band>`): good ≤ 60 ms (green text), fair 61–150 ms (amber text), poor > 150 ms (orange text) — a secondary quality cue. Each band's text clears AA on the chip tint (the conservative text-on-opaque bound, [COLOR §4.4]).

Tokens consumed: `comp.statusChip.bg` (state tint surface), `comp.statusChip.border`, `comp.statusChip.dot.<state>` (= `state.<state>.fill`), `comp.statusChip.label` (= `state.<state>.onTint text/icon`, [COLOR §3.1]), `comp.statusChip.pathBadge.{direct, relay}. {bg, label}`, `comp.statusChip.rtt.{good, fair, poor}`, `comp.statusChip.padX` (= `space.scale.3`), `...gap` (= `space.scale.2`), `comp.statusChip.radius` (= `radius.scale.full`, a pill).

Accessibility. Role: status (a polite live-region — rtt/path changes are announced without stealing focus). Accessible name composes "WireGuard, direct path, 28 milliseconds, good". The dot is decorative; the meaning is in the label. Not interactive by default (a status, not a button); when the Client makes it tappable ($\rightarrow$ open connection details), it becomes role button with a 44 px target. rtt text never relies on colour alone — the band is reinforced by the numeric value.

Light + dark. The tint (bg) and dot/label swap via the semantic fork; on dark the tint is a low-alpha mix over #151920 and the text uses the bright state stops ([COLOR §3.1, §4.4]). path-badge sub-pill contrast is proven in both themes.

No-overlap guarantee. Single-row flex:

`dot | icon | label | pathBadge | spacer | rtt`. The label truncates with ellipsis before it can collide with the path-badge; the spacer (flex-grow) keeps rtt right-aligned with $\geq \text{space.scale.2}$ from the path-badge so the two never touch. The pulsing reconnecting dot scales within its own box (it never grows over the icon).

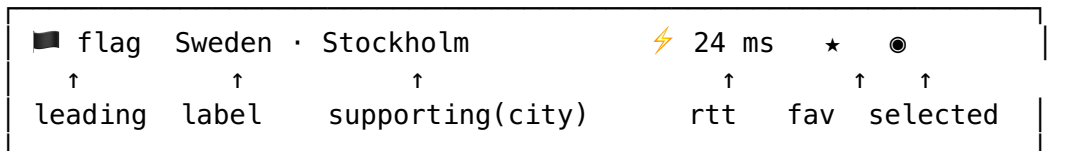
4. Exit & server family

The three components that let the user pick where their traffic exits. Built from `ExitOption { id, kind, label, country, rtt_ms, jurisdiction }` [FFI §2].

4.1 ServerListItem

Purpose. One selectable exit/server row: country, latency, kind, multi-hop jurisdiction label, favourite toggle. The atom of `ServerList/ExitPicker`.

Anatomy.



`comp.serverListItem.bg.<state> / .border`

Parts: `leading` (country flag / kind glyph), `label` (country / exit label), `supporting` (city or jurisdiction), `rtt` (latency + quality colour band, same bands as §3), `kindBadge` ("privacy_exit" shield vs "network" globe), favourite (star toggle), `selectedMark` (radio dot / check), `bg`, `border`.

States. `default` · `hover` · `pressed` · `selected` (active exit) · `focused` (keyboard) · `disabled` (offline/over-capacity) · `favourite` (starred) · `loading-rtt` (rtt skeleton while probing). `selected` paints `surface.brandTint bg` + a leading check + the per-app accent ([COLOR §6]).

Tokens consumed: `comp.serverListItem.bg`.
`{default,hover,pressed,selected,disabled}`,
`comp.serverListItem.label`, `comp.serverListItem.supporting`,
`comp.serverListItem.rtt.{good,fair,poor}`,
`comp.serverListItem.kindBadge.*`, `comp.serverListItem.favourite`.
`{on,off}`, `comp.serverListItem.selectedMark`,
`comp.serverListItem.border`, `comp.serverListItem.padX/padY` (scale.3/
scale.2), `comp.serverListItem.height` (≥48 px, ≥44 tap).

Accessibility. Role: option inside the listbox of `ServerList` (single-select). Accessible name: "Sweden, Stockholm, privacy exit, 24 milliseconds, good latency, favourite, selected". The favourite star is a **nested** button ("Add to favourites" / "Remove from favourites") with its own 44 px target, reachable by tab and not part of the row's primary activation. Arrow keys move between options; Enter/Space selects. Selected state is conveyed by the check glyph + `aria-selected`, not colour alone.

Light + dark. Row `bg/selected-tint/border` swap via the semantic fork; the flag image is theme-neutral; `rtt` band text uses the proven on-surface stops per theme.

No-overlap guarantee. A grid row: [`leading`] [`label/supporting` stack] [`rtt`] [`kindBadge`] [`favourite`] [`selectedMark`], each in its own cell with min-width. The `label/supporting` stack flex-grows and ellipsises; `rtt/badges/star` are fixed-width and right-anchored with `space.scale.2` gaps — no cell overlaps. The favourite star's 44 px tap box does not overlap the row's tap box logic (the star captures its own taps; the rest selects the row).

4.2 ServerList

Purpose. The scrollable, groupable, filterable list of `ServerListItems` — the body of the exit picker and the Console's location browser.

Anatomy. header (optional group label, e.g. "Recommended", "All countries"), `searchField` (a `TextField`, §11.2, when filterable), rows (virtualised `ServerListItems`), `groupHeaders` (sticky country/region dividers), `scrollbar`, `emptyState` (§11.16), `skeleton` (§11.17 while loading).

States. `populated` · `filtered` (search active, count shown) · `empty` (no servers) · `empty-filtered` (no match → `EmptyState` with "Clear filter") · `loading` (skeleton rows) · `error` (load failed → Banner + retry).

Tokens consumed: `comp.serverList.bg` (= `surface.default`), `comp.serverList.groupHeader.{bg,label}`, `comp.serverList.divider` (= `border.default`), `comp.serverList.scrollThumb`, plus the `comp.serverListItem.*` and `comp.field.*` (search) and `comp.skeleton.*` it composes.

Accessibility. Role: `listbox` (single-select) wrapping the option rows; group headers use `group` + `aria-label`. The search field has role `searchbox` and updates an `aria-live` count ("12 of 240 servers"). Keyboard: type-to-search, arrow to move, Home/End to jump, PageUp/Down to page. Virtualised rows keep `aria-setsize`/`aria-posinset` accurate so screen readers announce "row 4 of 240" despite virtualisation.

Light + dark. Surface, dividers, sticky-header bg swap via the fork; sticky headers get a `surface.raised` tint so they separate from scrolling rows in both themes.

No-overlap guarantee. Sticky group headers reserve their own height band; a scrolling row never renders under a sticky header (the header has an opaque `surface.raised` bg + `elevation.card` on dark so content does not bleed through). The scrollbar thumb is inset and never overlaps the `rtt`/favourite trailing slot.

4.3 ExitPicker

Purpose. The full exit-selection surface: a launcher (current exit summary) that opens a `ServerList`, supports search, favourites, recents, and a **multi-hop chain** (delegating to §6). The Client's "choose where I exit" control.

Anatomy. `trigger` (current-exit summary chip: flag + label + chevron), `panel` (a Sheet on Client / a Popover on desktop containing the `ServerList`), tabs (Recommended / Favourites / All / Multi-hop, §11.9), `chainSummary` (when a multi-hop chain is active, shows the hop sequence, §6).

States. closed (showing the trigger summary) · open · searching · multihop-chain (chain editor tab active) · selected (a new exit chosen, panel closes, trigger updates with a 180 ms cross-fade).

Tokens consumed: `comp.exitPicker.trigger.{bg, label, chevron}`, `comp.exitPicker.panel.{bg, radius}` (= `surface.raised`, `radius.scale.lg`), `comp.exitPicker.row.selected.bg` (= `surface.brandTint`), plus `comp.tabs.*`, `comp.serverList.*`, `comp.dialog.*` (the panel is a Sheet/Popover, §11.6).

Accessibility. The trigger is role button with `aria-haspopup="listbox"` and `aria-expanded`. Opening moves focus into the search field; Esc closes and returns focus to the trigger (focus restoration). The active exit is announced on selection ("Exit set to Sweden, Stockholm"). On Client (Sheet) the open panel traps focus; on desktop (Popover) focus is trapped until Esc/outside-click.

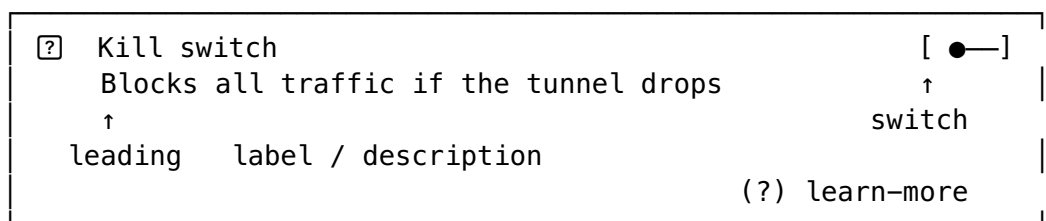
Light + dark. Trigger and panel surfaces swap via the fork; the panel uses `elevation.menu/modal` shadows that darken on dark ([DT §6.3]).

No-overlap guarantee. The panel opens **above** an `overlay.scrim` (Client Sheet) or as an opaque popover with `elevation.menu` (desktop) — never a translucent layer over the list. The trigger summary truncates the exit label before it can collide with the chevron. The chain summary (multi-hop) lays hops in a horizontal scroll row with arrow separators in their own boxes (§6).

5. ShieldToggle

Purpose. The protection-feature toggles driven by the Shields DTO [FFI §2]: **kill-switch**, **DNS protection**, **DAITA** (defence against AI traffic analysis), **post-quantum** (and per-route split-tunnel as a list, which uses `ServerListItem`-class rows). Each is a labelled row with a Switch (§11.3), a description, and a "what is this" affordance.

Anatomy.



`comp.shieldToggle.bg / .border`

Parts: leading (feature glyph), label (feature name), description (one-line explainer), switch (the Toggle, §11.3), learnMore (info button → Tooltip/Sheet), lockIcon (when locked by org policy), bg, border.

States. off · on · pending (awaiting set_shields round-trip — switch shows a small spinner, aria-busy) · disabled (unavailable on this platform/topology) · locked-by-policy (Console/Connector pushed a forced value; switch is non-interactive + a lock glyph + "Managed by your administrator").

Kill-switch cross-link. When kill-switch is on and the tunnel enters Danger/Down, the **Danger banner** (§11.8) appears; the ShieldToggle row for kill-switch shows an active (green) state to reassure the user the block is working. These two components are coordinated by the screen, not by overlap.

Tokens consumed: comp.shieldToggle.bg.{default, locked}, comp.shieldToggle.label, comp.shieldToggle.description (= text.secondary), comp.shieldToggle.leading.{off, on}, comp.shieldToggle.lockIcon (= text.tertiary), plus comp.switch.* (the toggle) and comp.tooltip.* (learn-more).

Accessibility. Role: switch (the toggle owns the state) with the row label as its accessible name and the description as aria-describedby. pending sets aria-busy; locked-by-policy sets aria-disabled + appends "managed by administrator" to the name. The learn-more is a separate button ("About kill switch"). Toggling announces "Kill switch, on" / "off". 44 px target on the switch; the whole row is also tappable as a larger target that forwards to the switch.

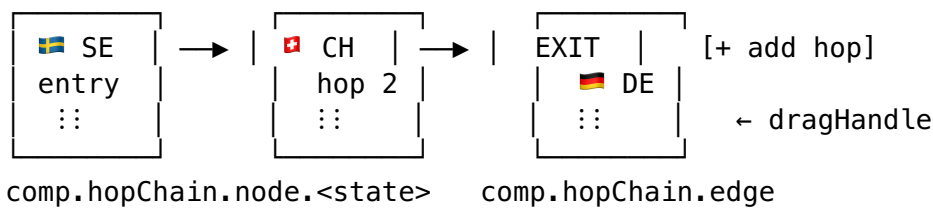
Light + dark. Row bg/label/description and the switch track/thumb swap via the fork (the on track uses action.primary per app accent, the off track uses border.strong). Lock glyph uses text.tertiary (proven AA in both).

No-overlap guarantee. Grid: [leading] [label/description stack] [learnMore] [switch]. The description wraps to at most 2 lines within its cell and never flows under the switch; the switch is fixed-width, right-anchored with space.scale.3. The lock glyph (when present) replaces the learnMore slot — they never both occupy it.

6. MultiHopChainEditor

Purpose. Build and reorder a **multi-hop exit chain** (entry → ... → exit), each hop an ExitOption with a jurisdiction label [FFI §2]. The advanced Client surface (and a Console policy primitive) for "route through Sweden then Switzerland".

Anatomy.



Parts: node (a hop card: flag + country + jurisdiction + role tag entry/middle/exit + drag handle + remove), edge (the directional arrow/connector between hops), addHop (button to append), dragHandle, removeHop, invalidMark (when a chain is invalid).

States. 1-hop (single exit, edges hidden) · 2-hop/n-hop · add-hop (a placeholder slot + ServerList to pick) · reorder (a node lifted, others shift, drop target highlighted) · invalid-chain (e.g. duplicate hop, unreachable pair, jurisdiction-conflict → node + edge paint feedback.error, the apply button disables, a Banner explains).

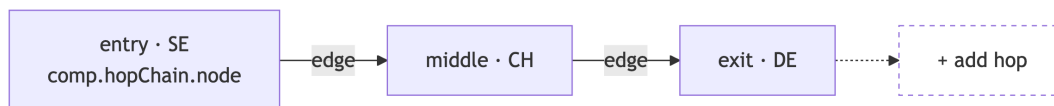
Tokens consumed: comp.hopChain.node.

{default, selected, dragging, invalid}. {bg, border, label},
comp.hopChain.edge. {default, invalid} (= border.strong /
feedback.error), comp.hopChain.roleTag. {entry, middle, exit},
comp.hopChain.dragHandle, comp.hopChain.addHop.* (= comp.button.secondary.*), plus comp.serverList.* (hop picker),
comp.banner.* (invalid explainer).

Accessibility. The chain is an ordered list; each node is a listitem with role-tag in its name ("Hop 1 of 3, entry, Sweden"). Reordering is **keyboard-operable**: focus a node, Space to "lift", arrow keys to move, Space to drop (a documented alternative to drag, WAI APG); a live-region announces "Hop 2 moved to position 1". Drag is pointer-only sugar; the keyboard path is the canonical one. Remove is a nested button ("Remove hop 2"). Invalid chains set aria-invalid on the offending node and link the Banner via aria-describedby.

Light + dark. Node cards use surface.raised; edges use border.strong; invalid nodes/edges use feedback.error — all swap via the fork. The lifted (dragging) node gets elevation.menu which darkens on dark ([DT §6.3]).

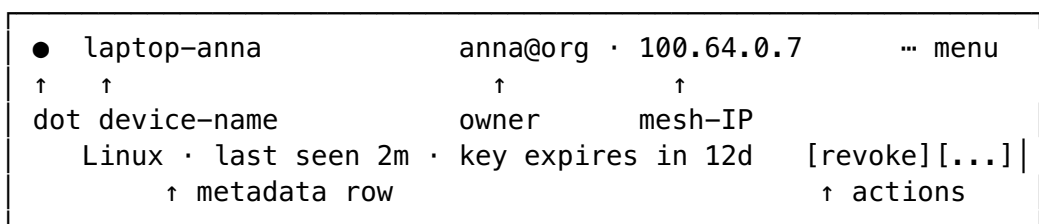
No-overlap guarantee. Nodes are laid in a horizontal flex (wrapping on narrow Client / single-row scroll on desktop) with the edge arrow in its **own** fixed box between nodes — the arrow never overlaps a node's flag or label. During reorder the lifted node is rendered on an elevated layer but the drop gap is reserved by shifting siblings (no overlap with the resting nodes). The remove and drag-handle controls sit in dedicated corners of each node card with their own padding.



7. PeerCard / DeviceListItem

Purpose. Represent one **device** / **peer** in the mesh — used by the Console (admin device roster) and the Connector (appliance's known peers). Shows identity, online state, key health, advertised routes, and admin actions.

Anatomy.



comp.peerCard.bg.<state> / .border

Parts: statusDot (online state), name (device name), owner (participant handle, §11.4.104), meshIp, meta (OS · last-seen · key-expiry · advertised routes), tags (route/tag chips, Badge §11.13), actions (approve / revoke / rename / expire-key — a menu + inline buttons), bg, border.

States. online (green dot) · idle (amber dot, seen recently) · offline (grey dot) · expiring-key (amber warning chip — key expires soon) · revoked (struck-through, red, read-only) · pending-approval (new device awaiting admin approve/deny — primary "Approve" + "Deny" actions). online/idle/offline reuse the connection-state-family colours but are **device-presence**, not tunnel state (a distinct semantic, [COLOR §2]/§8 D-CL-2).

Tokens consumed: comp.peerCard.bg.

{default, selected, revoked, pending}, comp.peerCard.statusDot.
{online, idle, offline}, comp.peerCard.name, comp.peerCard.owner (= text.secondary), comp.peerCard.meta (= text.tertiary),
comp.peerCard.expiryWarn (= feedback.warning), comp.peerCard.revoked (= feedback.error + strikethrough), plus comp.badge.* (tags),
comp.button.* (actions), comp.card.* (container), comp.menu.*.

Accessibility. Role: listitem (in the roster list) or article when standalone. Accessible name: "laptop-anna, owned by anna, online, key expires in 12 days". The status dot is decorative; presence is in the name. Actions are a menu button ("Device actions for laptop-anna") + inline

buttons for the high-priority pending-approval/revoke paths. revoked rows set aria-disabled and are non-interactive except "view". Destructive actions (revoke, expire-key) open a confirm Dialog (§11.6).

Light + dark. Card surface, dots, meta text, warning/error accents swap via the fork; the revoked strikethrough colour reads in both themes.

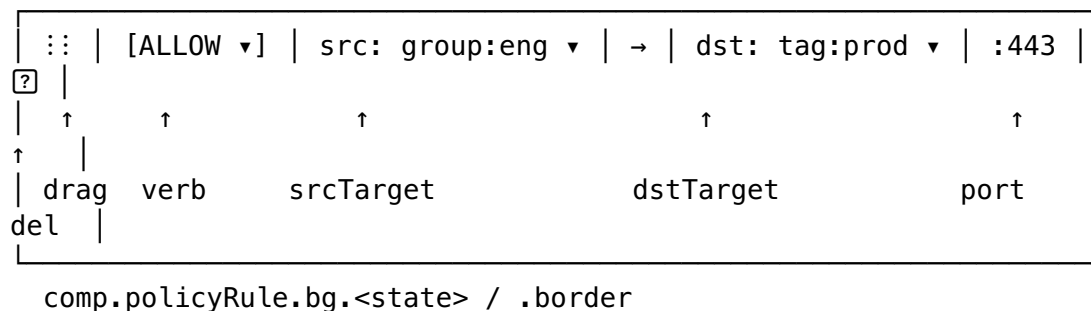
No-overlap guarantee. Two-row grid: row 1 [dot] [name] [owner] [meshIp] [menu], row 2 [meta...] [tags] [actions]. Each cell has min-width; long device names / owner handles ellipsis within their cell and expose the full value via tooltip + accessible name. The trailing menu/actions are fixed-width and right-anchored with space.scale.2. The status dot reserves an 8 px box left of the name and never overlaps the glyph.

8. PolicyEditor primitives (Console)

Purpose. The building blocks of the Console ACL/policy editor: a **rule row**, an **allow/deny toggle**, and a **target picker** (source/destination/port). One rule = one row; many rows = a policy.

8.1 RuleRow

Anatomy.



Parts: dragHandle (reorder — order matters in first-match ACLs), verb (allow/deny toggle, §8.2), srcTarget/dstTarget (target pickers, §8.3), portField (port/range), arrow (src→dst connector), delete, bg, border, conflictMark.

States. allow (verb green-tinted) · deny (verb red-tinted) · draft (unsaved — dashed border + "unsaved" badge) · conflict (shadowed by an earlier rule, or contradicts another → amber conflictMark + a Tooltip explaining which rule shadows it) · disabled (rule toggled off, greyed) · selected.

Tokens consumed: comp.policyRule.bg.

{default, draft, conflict, disabled, selected}, comp.policyRule.border.
{default, draft, conflict}, comp.policyRule.arrow,

`comp.policyRule.conflictMark (= feedback.warning)`, plus
`comp.policyVerb.* (§8.2)`, `comp.select.* (target pickers, §8.3)`,
`comp.field.* (port)`, `comp.button.* (delete)`.

8.2 AllowDenyToggle

A two-segment segmented control (Allow | Deny) — a specialised Tabs/segmented control. allow selected → green segment + check glyph; deny selected → red segment + block glyph. Tokens: `comp.policyVerb`.
`{allow, deny}`. `{bg, label}`. `{selected, unselected}`
(`allow=feedback.success`, `deny=feedback.error`, [COLOR §2.4]). Role: radiogroup of two radios; the selected verb is in the rule's accessible name ("Allow rule"). Colour-independent: glyph (check / block) reinforces the verb.

8.3 TargetPicker

A typed Select (§11.5) constrained to policy targets: `user:`, `group:`, `tag:`, `cidr:`, `host:`, `*` (any). Anatomy: `prefixBadge (the kind)`, `value (autocomplete)`, `clear`. Validates the value against the kind (a `cidr:` must be a valid CIDR → `aria-invalid + feedback.error` on bad input). Tokens: `comp.targetPicker.prefixBadge.<kind>`, `comp.targetPicker.value`, plus `comp.select.*`. Role: combobox with an autocomplete listbox; the kind is in the accessible name ("destination, tag, prod").

Accessibility (PolicyEditor overall). The policy is an ordered list of `RuleRow` listitems (order = precedence, announced "rule 3 of 8"). Keyboard reorder per §6 pattern. Conflicts set `aria-invalid + link` the explainer Tooltip. The arrow → between `src` and `dst` is decorative (`aria-hidden`); the relationship is in the name ("Allow from group eng to tag prod on port 443").

Light + dark. Row/verb/target surfaces and the allow-green / deny-red / conflict-amber accents swap via the fork. Draft dashed border reads in both themes.

No-overlap guarantee. A single grid track per row: every segment (`drag | verb | src | arrow | dst | port | delete`) is a fixed/min-width cell; long target values ellipsize and expose full value via tooltip — they never overflow into the adjacent cell. On narrow Console widths the row wraps to two grid lines (`src→dst` on line 1, `port+actions` on line 2) rather than overlapping. The arrow sits in its own 24 px cell.

9. TopologyGraph (Console)

Purpose. A node/edge visualisation of the mesh: devices/exits as **nodes**, connections as **edges** (direct vs relay vs blocked), with health colouring. The Console's "see my network" surface.

Anatomy. node (a peer/exit: glyph + label + health ring), edge (a line between nodes: solid=direct, dashed=relay, red=blocked), nodeLabel, healthRing, legend, controls (zoom / fit / filter), selectionHalo, tooltip (hover detail → §11.14).

States.

- **node:** healthy (green ring) · degraded (amber ring) · down (orange ring + dimmed) · selected (brand halo) · focused (keyboard ring).
- **edge:** direct (green solid) · relay (teal dashed) · blocked (red, with a block glyph at midpoint) · highlighted (when an incident node is selected, its edges thicken).

Tokens consumed: comp.topology.node.

{healthy, degraded, down, selected}. {fill, ring, label},
comp.topology.edge. {direct, relay, blocked, highlighted},
comp.topology.selectionHalo (= border.focus), comp.topology.legend.*,
comp.topology.canvas (= surface.sunken), plus comp.tooltip.*.

Accessibility. A graph is the hardest a11y surface; HelixVPN ships **dual representation:** the visual canvas (role img with a rich aria-label summarising "12 nodes, 2 degraded, 1 blocked edge") **plus** an equivalent, always-available **data table / tree** (the same nodes+edges as a navigable treegrid) so a screen-reader / keyboard user has a non-visual path — the visual graph is never the only way to read the topology (§11.4.117 pixel-only surfaces get a structured fallback). Nodes are keyboard-focusable in a deterministic order; Enter selects (opens the PeerCard, §7); edges are reachable via their endpoints. Health is conveyed by ring **shape pattern** (solid/dashed/dotted) + label, not colour alone.

Light + dark. Canvas (surface.sunken), node fills, edge colours, and rings swap via the fork. Edge **line styles** (solid/dashed) are theme-invariant so the direct/relay/blocked distinction survives in both themes regardless of colour.

No-overlap guarantee. A force-directed/elk layout enforces a minimum node-to-node distance ($\geq$ node-diameter + space.scale.4) and routes edges to avoid crossing node boxes; node labels are placed in a non-overlapping band (below the node, collision-resolved — a label that would collide is offset or truncated + tooltip). The legend and controls are in fixed corners on an opaque surface.raised panel, never over a node. UNVERIFIED — the

exact layout engine (elk vs d3-force) and its label-collision parameters are pinned by the Console graph spec ([screens-console.md]); marked UNVERIFIED until that spec exists.

10. AuditLogRow (Console)

Purpose. One entry in the Console audit/event log: timestamp, actor, action, target, severity, outcome — expandable to the full event payload. The forensics/compliance surface (composes the §11.4.83 evidence model).

Anatomy.

▶	14:02:11Z	●	anna	revoked device	laptop-bob
success					
↑	↑	↑	↑	↑	↑
exp	time	sev	actor	action	target
outcome	(expanded) full JSON payload · source IP · request id				

comp.auditRow.bg.<state> / .border

Parts: expandToggle (chevron), time (mono, UTC), severityDot, actor (participant handle), action (verb), target, outcome (success/fail badge), payload (collapsible monospace detail), bg, border.

States. Severity: info · success · warn · error · security (a distinct purple/violet accent for security-relevant events). Interaction: default · hover · selected · expanded (payload shown).

Tokens consumed: comp.auditRow.bg.{default,hover,selected}, comp.auditRow.severityDot.{info,success,warn,error,security} (= feedback.{info,success,warning,error} + accent.security), comp.auditRow.time (= text.secondary, font.mono), comp.auditRow.actor, comp.auditRow.action (= text.primary), comp.auditRow.outcome.{success,fail}, comp.auditRow.payload.{bg,text} (= surface.sunken, font.mono), plus comp.badge.*(outcome).

Accessibility. Role: row in a table/grid (columns: time, severity, actor, action, target, outcome) — proper table semantics so screen readers announce column headers. The expand toggle is button with aria-expanded/aria-controls pointing at the payload region. Severity is in the row's accessible name + a glyph, not colour alone ("error: revoke failed"). Time is announced in a readable form. The monospace payload is a region with a copy-to-clipboard button.

Light + dark. Row/payload surfaces, severity dots, outcome badges swap via the fork; the monospace payload uses `surface.sunken` (#EDEF3 light / #0B0E13 dark, [COLOR §2.1]) so code reads in both themes.

No-overlap guarantee. Fixed table grid: each column is a sized cell; action and target ellipsis within their cells (full value in the cell's tooltip + accessible name). The expanded payload renders **below** the row in its own full-width region pushing subsequent rows down (it never overlays the next row). The severity dot reserves its column; the outcome badge is right-anchored.

11. Foundational components

The platform-neutral primitives every app composes. Each ships light+dark, the §1.2 interaction model, the §1.3 a11y baseline, and the §1.5 no-overlap guarantee; sections below add only what is component-specific.

11.1 Button family

Purpose. The four button intents: **primary** (filled brand action), **secondary** (tonal/outline), **text** (low-emphasis), **danger** (destructive filled red).

Anatomy. `container` (fill/outline), `label`, `leadingIcon/trailingIcon` (optional), `spinner` (loading), `focusRing`, `overlay.<hover|pressed>`.

Variants × states.

Variant	bg (rest)	label	border	Use
primary	<code>action.primary</code> (per-app accent)	<code>text.onBrand</code> (white)	none	the main action
secondary	<code>surface.raised</code>	<code>action.primary</code>	<code>border.strong</code>	secondary action
text	transparent	<code>action.primary</code>	none	tertiary / inline
danger	<code>feedback.error</code>	<code>text.onState</code> (white)	none	delete/revoke

States (all variants): `enabled` · `hover` · `pressed` · `focus` · `disabled` (0.38 opacity) · `loading` (spinner replaces `leadingIcon`, label stays, `aria-busy`).

Tokens: `comp.button.<variant>.{bg,label,border}.`
`{rest,hover,pressed,disabled}, comp.button.<variant>.focusRing,`
`comp.button.radius (= radius.scale.md), comp.button.padX (=`
`space.scale.3/4), comp.button.gap, comp.button.minHeight (40 md / 48`
`lg).`

Accessibility. Role button; visible label is the name (icon-only buttons require `semanticLabel`). `loading` sets `aria-busy` and keeps the button width stable (the spinner occupies the icon slot, no layout shift). Danger buttons that perform irreversible actions are paired with a confirm Dialog (§11.6). 44 px target.

Light + dark. All four variants swap fills/labels/borders via the fork; danger red and the per-app accent are proven AA for their white labels in both themes ([COLOR §4.3, §6]).

No-overlap. Single row: `leadingIcon | label | trailingIcon` with `space.scale.1` gaps; the label truncates with ellipsis (full value in tooltip) before colliding with a trailing icon. The spinner replaces the leading icon **in-place** — it never overlays the label.

11.2 TextField & form fields

Purpose. Single/multi-line text input and the form-field shell (label, helper, error) wrapping any input.

Anatomy. label (above), container (input box), `leadingIcon/trailingIcon` (e.g. clear, reveal-password), input (the editable text), placeholder, helper (below), error (below, replaces helper), counter (optional), `focusRing`, border.

States. `empty · focused · filled · error` (red border + error text + `aria-invalid`) · `disabled · readonly · loading` (trailing spinner, e.g. async validation).

Tokens: `comp.field.`
`{bg,border,input,placeholder,label,helper}.<state>, comp.field.border.{rest,focus,error} (= border.strong/border.focus/feedback.error), comp.field.error (= feedback.error),`
`comp.field.radius (= radius.scale.sm), comp.field.padX/padY,`
`comp.field.minHeight (≥40, multiline grows).`

Accessibility. The label is programmatically associated (for/`aria-labelledby`); helper/error via `aria-describedby`; error sets `aria-invalid=true` and is announced politely. Placeholder is **never** the only label (it disappears on input). Password reveal is a button with `aria-pressed` ("Show password"). Required fields mark `aria-required`. 44 px min input height.

Light + dark. Box/border/text/placeholder swap via the fork; error red and focus ring proven in both themes. Placeholder uses `text.tertiary` (AA, not the disabled exempt stop).

No-overlap. label / input / helper are stacked rows (never overlapping); leading/trailing icons sit inside the input box with `space.scale.2` insets so they never overlay the text caret/selection; the error text replaces the helper in the same row slot (they never both render).

11.3 Toggle / Switch

Purpose. A binary on/off control (the atom inside `ShieldToggle §5`).

Anatomy. track, thumb, focusRing, optional onIcon/offIcon inside the thumb.

States. off · on · focus · disabled · pending (thumb shows a micro-spinner during async commit, `aria-busy`).

Tokens: `comp.switch.track.{off,on,disabled}` (on = `action.primary`, off = `border.strong`), `comp.switch.thumb.{off,on,disabled}`, `comp.switch.focusRing`, `comp.switch.width/height` (≥ 44 px tap via padding).

Accessibility. Role switch, `aria-checked`. The thumb travels with a 180 ms motion ($\rightarrow$ instant on reduce-motion). State is conveyed by thumb position **and** optional on/off glyph, not colour alone. 44 px target.

Light + dark. Track/thumb swap via the fork; the on accent differs per app ([`COLOR §6`]).

No-overlap. Thumb travels **within** the track bounds; the optional glyph is centred in the thumb and never extends past it.

11.4 Checkbox & Radio

Purpose. Multi-select (Checkbox, with indeterminate) and single-select (Radio) controls.

Anatomy. box/circle, mark (check / dash / dot), label, focusRing.

States. Checkbox: unchecked · checked · indeterminate · focus · disabled.
Radio: unselected · selected · focus · disabled.

Tokens: `comp.checkbox.{box,mark}.`
`{unchecked,checked,indeterminate,disabled}, comp.radio.`
`{circle,dot}.``{unselected,selected,disabled}, ...focusRing,`
`comp.checkbox.radius (= radius.scale.sm); checked/selected fill =`
`action.primary, mark = text.onBrand.`

Accessibility. Roles `checkbox` (with `aria-checked="mixed"` for `indeterminate`) / `radio` (in a `radiogroup`). The visible label is the name and is itself clickable (extends the 44 px target around the 20 px glyph). Arrow keys move within a `radiogroup`; Space toggles a checkbox.

Light + dark. Box/circle borders, fill, and mark swap via the fork; the 20 px glyph keeps a 44 px transparent tap box in both.

No-overlap. `glyph | label` row with `space.scale.2` gap; the label wraps within its cell and never flows under the glyph; the focus ring offsets outside the glyph box.

11.5 Dropdown / Select

Purpose. Choose one (or many) from a list via a popover listbox. Underpins `ExitPicker` (§4.3) and `TargetPicker` (§8.3).

Anatomy. `trigger` (value + chevron), `popover` (surface + options), `option` (label + `leadingIcon` + `selectedCheck`), `search` (optional combobox input), `focusRing`.

States. `closed · open · option.{default,hover,selected,disabled} · focus · loading-options · error.`

Tokens: `comp.select.trigger.{bg,label,chevron,border}.<state>`,
`comp.select.popover.{bg,radius} (= surface.raised, radius.scale.md,`
`elevation.menu), comp.select.option.{bg,label}.`
`{default,hover,selected,disabled}, comp.select.selectedCheck, plus`
`comp.field.*` (when searchable).

Accessibility. Trigger button + `aria-haspopup="listbox" + aria-expanded`; popover listbox, options option with `aria-selected`. Esc closes + restores focus to trigger; type-ahead; arrow navigation with wrap; Home/End. Searchable variant is a combobox. 44 px option rows.

Light + dark. Trigger + popover surfaces and option hover/selected swap via the fork; the menu shadow darkens on dark ([DT §6.3]).

No-overlap. The popover is opaque (`surface.raised + shadow`), never a translucent wash; it is positioned to avoid covering the trigger (flips above when no room below). Option `label | check` row: the check is right-anchored, the label ellipsises before it.

11.6 Dialog / Sheet / Modal

Purpose. A focused, blocking surface for confirms, forms, and detail. **Dialog** (centred, desktop/Console), **Sheet** (bottom-anchored, Client mobile), **Modal** (generic blocking). Same component, three presentations.

Anatomy. scrim (backdrop), surface (the panel), header (title + close), body (content), footer (actions), handle (sheet drag affordance), focusTrap.

States. enter (240 ms, [DT §6.4]) · shown · exit · variant: dialog (centred) / sheet (bottom, drag-dismiss) / fullscreen (Client small screens).

Tokens: comp.dialog.scrim (= overlay.scrim), comp.dialog.surface (= surface.raised, elevation.modal), comp.dialog.radius (= radius.scale.lg), comp.dialog.header.{bg,title}, comp.dialog.footer.bg, plus comp.button.* (actions). z-index z.semantic.modal (1100, [DT §6.5]).

Accessibility. Role dialog + aria-modal=true + aria-labelledby (title)

- aria-describedby (body). **Focus trap** — tab cycles within; opening moves focus to the first focusable (or the panel); Esc closes (unless a forced confirm); closing **restores focus** to the trigger. The scrim click closes dismissible dialogs. Destructive confirms put the danger button as a distinct, non-default action.

Light + dark. Surface + scrim swap via the fork; the modal shadow is far darker on dark ([DT §6.3]); the scrim alpha differs per theme ([COLOR §2.3]).

No-overlap. The surface sits on an opaque-enough scrim so its labels meet contrast (white-on-scrim 14.68, [COLOR §4]); the header/body/footer are stacked regions (close button in the header corner, never over the title); the sheet drag handle is in its own top band above the header.

11.7 Snackbar / Toast

Purpose. Transient, non-blocking feedback ("Exit set to Sweden", "Copied"). Auto-dismisses; optionally one action.

Anatomy. container, icon (severity), message, action (one text button), dismiss (optional), progress (auto-dismiss timer, optional).

States. Severity: info · success · warn · error. Lifecycle: entering · shown · leaving. (error toasts get a longer/no auto-dismiss.)

Tokens: `comp.toast.{bg,message,icon}.<severity>`, `comp.toast.action` (= `action.primary` or `text.onState`), `comp.toast.radius` (= `radius.scale.md`, `elevation.menu`). z-index `z.semantic.toast` (1200, above modal so a confirm's result toast is visible, [DT §6.5]).

Accessibility. Role status (info/success → polite) or alert (warn/error → assertive). Auto-dismiss **pauses on hover/focus** and is long enough to read (min 5 s + action time); a toast with an action does not auto-dismiss until the action is reachable. Never the **only** channel for critical info (the Danger banner §11.8, not a toast, owns kill-switch/leak — a toast can be missed).

Light + dark. Severity bg/icon/message swap via the fork.

No-overlap. Toasts **stack** with `space.scale.2` gaps (newest at the stack origin), never overlapping; a max visible count caps the stack (older collapse). `icon | message | action | dismiss` single row; message ellipsises to 2 lines max before the action.

11.8 Banner / Alert (incl. the Danger banner)

Purpose. A persistent, in-flow (or top-pinned) message of higher weight than a toast: info/success/warning/error and the load-bearing **Danger banner** (leak / kill-switch tripped — the `Danger{kind}` state's full-width alarm).

Anatomy. `container`, `icon` (severity), `title`, `message`, `actions` (0–2 buttons), `dismiss` (info/success only — danger/error are not dismissible while active), `border/accentBar` (a leading colour bar).

States. `info · success · warning · error · danger`.

Variant	bg	accent / icon	dismissible?	z
info	<code>feedback.info tint</code>	blue	yes	in-flow
success	<code>feedback.success tint</code>	green	yes	in-flow
warning	<code>feedback.warning tint</code>	amber	yes	in-flow
error	<code>feedback.error tint</code>	red	no (while active)	in-flow
danger	<code>state.danger.fill</code> (solid red)	<code>text.onState</code> white, shield-alert	no	<code>z.semantic.danger</code> (2000 — top of everything, [DT §6.5])

The Danger banner is special: solid red (not a tint), full-width, pinned, and on the **highest** z-layer so **nothing** (modal, toast) can occlude it ([COLOR §3.2], [DT §6.5]) — mirroring the FFI rule that Danger overrides all intent [FFI §3.3]. It states the kind ("Traffic leak detected" / "Kill switch active — traffic blocked") and offers the safe actions (Reconnect / Stop).

Tokens: `comp.banner.{bg,title,message,icon,accentBar}.<variant>`, `comp.banner.dangerBg` (= `state.danger.fill`, `solid`), `comp.banner.radius` (= `radius.scale.md`, `danger = 0` for full-bleed top pin), plus `comp.button.*`.

Accessibility. Role alert for warning/error/danger (assertive — announced immediately); status for info/success. The Danger banner additionally moves focus to its primary safe action so a keyboard user can act at once. It is never auto-dismissed; it clears only when the underlying state leaves Danger. Icon + title carry the meaning (not colour alone).

Light + dark. Tints/accents swap via the fork; the **danger** solid red and white label are proven AA in both themes ([COLOR §4.3]) — danger does not change between themes (always the alarm).

No-overlap. Full-width stacked region:

`[accentBar][icon][title/message stack][actions][dismiss]`; the message wraps within its cell. Because the Danger banner owns z danger (2000) it is drawn **last/over** content but content is laid out **below** it (it pins to the top and pushes/insets content, not overlays it) — its own label is never occluded and it never hides an interactive element the user needs (the safe actions are inside the banner).

11.9 Tabs

Purpose. Switch between sibling views (ExitPicker tabs §4.3, settings sections). Also the base of the segmented AllowDenyToggle (§8.2).

Anatomy. `tablist`, `tab` (label + optional icon/badge), `indicator` (active underline/pill), `panel`.

States. `tab.{default,selected,hover,focus,disabled}`. Variant: underline (default) / segmented (pill, equal-width) / scrollable (many tabs).

Tokens: `comp.tabs.tab.{label,bg}.{default,selected,hover,disabled}`, `comp.tabs.indicator` (= `action.primary`), `comp.tabs.scrollFade`, `comp.tabs.minHeight` (≥44).

Accessibility. `tablist` / `tab` (aria-selected) / `tabpanel` (aria-labelledby). Arrow keys move between tabs; Tab moves into the panel. Selected is shown by indicator + aria-selected, not colour alone. Scrollable tabs keep the selected tab in view.

Light + dark. Tab labels, indicator, and hover swap via the fork.

No-overlap. The indicator animates **under** the tab labels (a separate layer, never over the text); scrollable tabs add a `scrollFade` gradient at the edge so a tab is never half-clipped ambiguously; tabs keep `space.scale.3` gaps.

11.10 NavRail / BottomNav / Drawer

Purpose. Top-level navigation — three responsive presentations of the same destination set: **BottomNav** (Client phone), **NavRail** (tablet/desktop side), **Drawer** (overflow / Console sections).

Anatomy. item (icon + label + optional badge), indicator (active), header/footer (logo, account), `expandToggle` (rail collapse), `scrim` (drawer).

States. item.`{default,selected,hover,focus}`; rail `collapsed/expanded`; drawer `open/closed`.

Tokens: `comp.nav.item.{icon,label,bg}.{default,selected,hover}`, `comp.nav.indicator (= action.primary)`, `comp.nav.surface (= surface.default/raised)`, `comp.nav.scrim (drawer = overlay.scrim)`, `comp.nav.badge.*`.

Accessibility. navigation landmark; items are link/button with `aria-current="page"` on the active one. BottomNav items are ≥ 44 px and labelled (icon + text). Drawer traps focus when modal, Esc closes, restores focus. Rail collapse toggle is button + `aria-expanded`; collapsed items keep their accessible name via tooltip.

Light + dark. Surfaces, item states, indicator swap via the fork; the drawer scrim alpha differs per theme.

No-overlap. BottomNav items are equal-width flex cells; the active indicator is a separate layer under the icon. Collapsed rail shows icon-only with the label in a tooltip (never a clipped label overlapping the next item). Drawer slides over a scrim (opaque-enough), content laid out beneath it.

11.11 Card

Purpose. A grouped surface for related content (the container of PeerCard §7, the hop nodes §6, dashboard tiles).

Anatomy. surface, border (optional), header/media/body/footer slots, overlay (when interactive).

States. `default (static)` · `interactive (hover/pressed when the whole card is a target)` · `selected` · `disabled`.

Tokens: `comp.card.surface` (= `surface.raised`), `comp.card.border` (= `border.default`), `comp.card.radius` (= `radius.scale.md`), `comp.card.elevation` (= `elevation.card`), `comp.card.overlay`. `{hover,pressed}`, `comp.card.selected.{bg,border}` (= `surface.brandTint` + `accent`).

Accessibility. A static card is a group/region (with `aria-label` when it has a heading). An interactive (whole-card-clickable) card is a single button/ link — and then it MUST NOT contain other independent interactive controls (nested-interactive is forbidden); if it needs sub-actions it stays a group with explicit buttons inside. 44 px when interactive.

Light + dark. Surface, border, elevation shadow swap via the fork (shadow darker on dark, [DT §6.3]); selected brand tint per theme.

No-overlap. Card slots are a vertical/grid layout with internal padding (`space.scale.4`); media is clipped to the card radius and never bleeds over the body text; an interactive overlay covers the whole card uniformly (it does not sit over only the label).

11.12 ListTile

Purpose. The generic one-line/two-line row (`leading` | `title/subtitle` | `trailing`) — the base many of the bespoke rows (`ServerListItem` §4.1, `ShieldToggle` §5) specialise.

Anatomy. `leading` (icon/avatar), `title`, `subtitle`, `trailing` (icon / switch / badge / chevron), `overlay`, `divider`.

States. `default` · `hover` · `pressed` · `selected` · `disabled`.

Tokens: `comp.listTile.{bg,title,subtitle,leading,trailing}.<state>`, `comp.listTile.divider` (= `border.default`), `comp.listTile.minHeight` (48 single / 64 two-line), `comp.listTile.padX` (= `space.scale.4`).

Accessibility. Role depends on use: `listitem` (in a list), `button` (actionable), `option` (selectable). Title is the name, subtitle is supporting. A trailing interactive control (switch/button) is a separate focusable target (nested), not folded into the row's activation. 44 px row.

Light + dark. All slots swap via the fork.

No-overlap. [`leading`] [`title/subtitle` stack] [`trailing`] grid; the text stack flex-grows + ellipsises; `leading/trailing` are fixed-width with `space.scale.2/4` gaps. Two-line subtitle wraps within its cell, never under the trailing control.

11.13 Badge

Purpose. A small status/count indicator — standalone (a tag chip) or anchored to a host (a notification count on a nav item).

Anatomy. container, label (text/count), optional dot (count-less presence).

Variants. `neutral · info · success · warning · error · count` (numeric, caps at "99+") · dot (small presence dot, no label).

Tokens: `comp.badge.{bg, label}.<variant>` (tints + on-tint text per `feedback.*`), `comp.badge.radius` (= `radius.scale.full`), `comp.badge.dot.<state>`.

Accessibility. A standalone status badge has its meaning in text (not colour alone). A count badge anchored to a host is exposed via the host's accessible name ("Notifications, 3 unread") — the badge itself is `aria-hidden` so it is not announced twice. A presence dot has a `semanticLabel` ("online").

Light + dark. Tints/labels swap via the fork; count badge on a coloured host keeps a contrasting fill.

No-overlap. An anchored badge reserves a corner box with its own padding and **offsets** the host's content so it never paints over the glyph/label it annotates; standalone badges keep `space.scale.1` internal padding so the label never touches the pill edge.

11.14 Tooltip

Purpose. A small, transient label revealing the name/help of a control (used by truncated labels, icon-only buttons, learn-more affordances §5).

Anatomy. bubble (surface + text), arrow (pointer to anchor).

States. `hidden · shown` (after a hover/focus delay; dismiss on blur/escape/move-away).

Tokens: `comp.tooltip.{bg, label}` (= `surface.inverse / text.onInverse` — a high-contrast inverse surface so the tip pops over any content), `comp.tooltip.radius` (= `radius.scale.sm, elevation.menu`), `comp.tooltip.maxWidth`.

Accessibility. The tip content is associated to the anchor via `aria-describedby` (supplementary) or, for an icon-only control with no visible label, **the tip IS the name** via `aria-label` (a tooltip alone must not be the only way to get a control's name for non-pointer users — so icon-only

controls carry the `aria-label` independently of hover). Keyboard: a tooltip shows on **focus**, not only hover; Esc dismisses; it never traps. It is not used for essential, action-only content (that is a Popover/Dialog).

Light + dark. Inverse surface flips so the tip stays high-contrast in both themes (dark tip on light app / light tip on dark app), UNVERIFIED exact inverse pairing pinned by `[color-system.md]`'s inverse-surface tokens (not yet enumerated there) — marked UNVERIFIED.

No-overlap. The bubble is opaque (`surface.inverse + shadow`), positioned to **not** cover its anchor (flips side when no room), and capped at `maxWidth` so it wraps rather than running off-screen over other content; the arrow sits in its own box pointing at the anchor edge.

11.15 ProgressIndicator (loading vs connecting)

Purpose. Two **distinct** progress affordances, kept separate per §11.4.107 (a "loading" spinner must not be confused with a "connecting" connection-state animation — they mean different things and a stuck spinner must be detectable):

1. **loading** — a generic indeterminate spinner / determinate bar for data fetches (a `ServerList` loading its options, a form submitting). Neutral colour (`action.primary` or `text.tertiary`).
2. **connecting** — the **connection-state** progress: the amber pulse that the `ConnectButton` (§2) and `StatusChip` (§3) show during Connecting/Handshaking/Reconnecting. It is the **state** animation, not a generic spinner, and is colour-bound to `state.connecting.fill`.

Anatomy. `track/arc` (spinner), `bar+fill` (determinate), `label` (optional "Loading..." / "Connecting...").

States. `loading.{indeterminate,determinate} · connecting (state pulse) · complete` (brief check before hide).

Tokens: `comp.progress.spinner` (= `action.primary` for loading), `comp.progress.bar.{track,fill}`, `comp.progress.connecting` (= `state.connecting.fill + motion.connectPulse`), `comp.progress.determinate.fill`.

Accessibility. `loading` → role `progressbar` (`aria-valuenow` for determinate; `aria-busy` on the host for indeterminate) + a visible/audible "Loading". `connecting` is **part of the `ConnectButton/StatusChip` semantics** (announced as "Connecting", §2/§3), not a separate progressbar. Indeterminate loading sets a timeout so a stuck spinner surfaces an error rather than spinning forever (§11.4.107 — a forever-spinner is a defect, not a state). Reduce-motion: the spinner becomes a static arc with `aria-busy`; the connecting pulse becomes static amber.

Light + dark. Loading spinner/bar use neutral+accent that swap via the fork; connecting uses the amber state stops per theme ([COLOR §3.1]).

No-overlap. A spinner replacing a button/icon occupies that slot exactly (no overlay onto the label, §11.1). A determinate bar is full-width in its own row; its label sits above/beside it, never over the fill.

11.16 EmptyState

Purpose. The "nothing here" placeholder for empty lists/searches/errors — an illustration/icon + headline + body + optional action.

Anatomy. illustration (icon/graphic), headline, body, action (0–1 buttons).

States. empty (no data yet) · empty-filtered (search/filter excludes all → "No matches" + "Clear filter") · error (load failed → error icon + "Retry").

Tokens: `comp.emptyState.{illustration,headline,body}` (text.secondary/ tertiary), `comp.emptyState.action` (= `comp.button.secondary.*`).

Accessibility. A status region announcing the empty/error message; the headline is a real heading; the action is a focusable button. Error variant uses role alert if it appears after a failed action. The illustration is decorative (aria-hidden) — the headline carries the meaning.

Light + dark. Illustration uses theme-aware line/fill tokens; text swaps via the fork.

No-overlap. Centred vertical stack (illustration / headline / body / action) with `space.scale.4` gaps; everything is in flow, nothing overlaps; on small Client screens the illustration scales down rather than overflowing.

11.17 Skeleton

Purpose. The shimmer placeholder shown while real content (server lists, peer rosters, audit logs) loads — distinct from the spinner: skeletons preview **layout**, the spinner shows **activity**.

Anatomy. block (a shaped placeholder: line / circle / rect matching the real element), shimmer (the animated sweep).

States. shimmer (animated) · static (reduce-motion → a solid muted block, no sweep).

Tokens: `comp.skeleton.base` (= `surface.sunken`),
`comp.skeleton.highlight` (the sweep, a lighter tint), `comp.skeleton.radius`
(matches the real element), `motion.semantic.skeletonSweep` (→
`reducedMotion` static).

Accessibility. The skeleton region sets `aria-busy=true` and a `aria-label` ("Loading servers"); individual blocks are `aria-hidden` (they are decorative placeholders). On `reduce-motion` the sweep stops (static block) — it never strobes. When real content arrives, focus and `aria-busy` hand off cleanly.

Light + dark. Base + highlight use `surface.sunken` + a lighter sweep per theme so the shimmer reads on both.

No-overlap. Skeleton blocks mirror the real layout's grid (same cells, same gaps) so there is no overlap and **no layout shift** when real content replaces them (the skeleton reserves the exact space the content will fill).

12. Surfaced decisions & cross-doc contracts

id	Decision / contract	Status
D-CL-1	Every component reads only <code>comp.*</code> tokens (which reference semantics); no component hard-codes a colour/length/duration. Enforced by the token tier lint [DT §9 CM-token-tiers-valid] + a component-source grep gate.	decided
D-CL-2	Device presence (online/idle/offline, PeerCard §7) reuses the connection-state colour <i>family</i> but is a distinct semantic from tunnel <code>TunnelStatus</code> — they must not be conflated in tokens (presence ≠ protection).	decided
D-CL-3	The <code>ConnectButton</code> (§2) is total over the 7-variant <code>ffi::TunnelStatus</code> [FFI §3.2]; a new FFI variant is a contract change requiring a new <code>comp.connectButton.bg.* slot</code> (gate CM-connectbutton-state-total).	contract
D-CL-4	Critical safety info (leak / kill-switch) is owned by the Danger banner (§11.8, z 2000), never a dismissible toast — a toast can be missed.	decided
D-CL-5	Whole-card / whole-row interactive surfaces MUST NOT nest independent interactive controls (nested-interactive ally rule); such rows stay group + explicit buttons.	decided
D-CL-6		decided

id	Decision / contract	Status
	TopologyGraph (§9) ships a structured non-visual fallback (treegrid) alongside the canvas — the graph is never the only way to read the mesh (§11.4.117).	
C-CL-A (consumes)	The <code>comp.<component>.<part>.<state></code> naming + tier rules are owned by [design-tokens.md §2, §7]; this catalog's token citations MUST resolve there.	contract
C-CL-B (consumes)	Colour values + contrast proofs for every slot are owned by [color-system.md §2–§4]; a11y floors here cite those ratios.	contract
C-CL-C (consumes)	The 7-variant <code>ffi::TunnelStatus + Shields/ExitOption</code> DTOs are owned by [FFI §2, §3.2]; <code>ConnectButton/StatusChip/ShieldToggle/ExitPicker</code> render them.	contract
C-CL-D (provides)	This catalog is the input contract for the visual-regression + golden-screenshot suite ([visual-regression-and-qa.md], §11.4.162/.168) and the screen-composition docs ([screens-client.md]/[screens-console.md]).	contract
U-CL-1 UNVERIFIED	TopologyGraph layout engine + label-collision params (§9) — pinned by [screens-console.md]; UNVERIFIED until that spec exists.	open
U-CL-2 UNVERIFIED	Tooltip inverse-surface token pairing (§11.14) — <code>surface.inverse/text.onInverse</code> not yet enumerated in [color-system.md]; UNVERIFIED until added there.	open
U-CL-3 UNVERIFIED	Exact pixel diameters/heights cited (<code>ConnectButton</code> 180/220 px, sizes table) are recommended defaults; final values pinned by [screens-client.md] + the <code>size.*</code> token tier; UNVERIFIED until that tier is enumerated.	open

Sources verified

- **The component catalog — anatomy, state sets, slots, variants, sizes, sketches, and the no-overlap / light+dark / a11y design for every component — NO external source needed — original HelixVPN design work**, modeled on the well-established headless-component / design-system component-catalog pattern.

- **Accessibility patterns (roles, keyboard interaction, focus management, listbox/combobox/dialog/switch/tabs/treegrid, drag-alternative keyboard reorder, name/description association, live-regions)** — W3C WAI-ARIA Authoring Practices Guide (APG) patterns and WCAG 2.1 success criteria (SC 1.4.3 contrast, SC 1.4.11 non-text contrast, SC 2.4.7 focus visible, SC 1.4.1 use-of-colour, SC 4.1.2 name-role-value), <https://www.w3.org/WAI/ARIA/apg/> and <https://www.w3.org/TR/WCAG21/> (verified 2026-06-25). The ≥44 px target floor follows WAI/Apple HIG/Material touch-target guidance.
- **comp.<component>.<part>.<state> token naming, the three-tier model, scales, motion/reduce-motion, z-index (danger 2000), elevation** — final/v10-design/design-tokens.md §2, §6, §7, §8 (sibling, this wave).
- **Colour roles, connection-state palette, per-app accent, contrast ratios, the colour-layer no-overlap rules** — final/v10-design/color-system.md §2–§6 (sibling, this wave).
- **7-variant ffi::TunnelStatus (Disconnected · Connecting · Handshaking · Connected{direct|relay} · Reconnecting · Down · Danger), Shields (kill_switch/dns_protection/daita/post_quantum/split_tunnel), ExitOption (id/kind/label/country/rtt_ms/jurisdiction)** — final/v04-client/ffi-surface.md §2, §3.2 (read 2026-06-25).
- **Client UI overview (ConnectButton/StatusChip/state-token usage), 3-app / 8-platform matrix, §11.4.162 OpenDesign design-system mandate (light+dark, no-overlap, visual regression)** — final/03-client-core-and-ui.md §7 + final/SPECIFICATION.md §3 + constitution §11.4.162 (read 2026-06-25).
- Items explicitly marked UNVERIFIED (U-CL-1 topology layout engine, U-CL-2 tooltip inverse-surface tokens, U-CL-3 exact pixel sizes) are pending their named sibling specs/token tiers per §11.4.6 — not asserted as fact.